

Kneser's conjecture: $G_{n,k}$ vertex set $[2n+k]^{(n)}$ Join A to B iff $A \cap B = \emptyset$. $\chi(G_{n,k}) = k+2$.

$(k+2)$ is possible: If $\max A \in \{2n, 2n+1, \dots, 2n+k\}$, then color A by $\max A$. Otherwise, color A by another class.

$(k+1)$ is not possible: Choose $(2n+k)$ pts (in a general position) on S^{k+1} .

For any point y on S^{k+1} , we consider the tuple $t_y = (\langle y, x_1 \rangle, \langle y, x_2 \rangle, \dots, \langle y, x_{2n+k} \rangle)$. For any $A \in [2n+k]^{(n)}$, we can identify it with a region in $\mathbb{R}^{2n+k}$, that is

$$V_A = \{(r_1, r_2, \dots, r_{2n+k}) : r_i > 0 \text{ when } i \in A\}.$$

If a $(k+1)$ -coloring for $G_{n,k}$ exists, say $[2n+k]^{(n)} = C_1 \sqcup C_2 \sqcup \dots \sqcup C_{k+1}$. We have the famous Borsuk-Ulam thm:

Thm: If $A_1, A_2, \dots, A_{n+1}$ cover S^n & all A_i are open/closed, then one of them contains a pair of antipodal points.

Back to the proof.

We can make $(k+1)$ open sets $A_i = \bigcup_{B \in C_i} A_B$, $A_B = \{y \in S^{k+1} : t_y \in V_B\}$.

And we can make one closed set $V = S^{k+1} \setminus \bigcup_{i=1}^{k+1} A_i$. Then $A_1, \dots, A_{k+1}, V$ cover S^{k+1} . By Borsuk-Ulam, one of them contains a pair of antipodal points. But it can not be A_i . If so, say $x, -x \in A_i$. So $x \in A_{B_1}$, $-x \in A_{B_2}$ for some $B_1, B_2 \in C_i$. By the assumption, $B_1 \cap B_2 \neq \emptyset \Rightarrow \exists j \in B_1 \cap B_2 \Rightarrow \langle x, x_j \rangle > 0$ & $\langle -x, x_j \rangle > 0$. $\times$
 $V = \{y \in S^{k+1} : \text{for at most } (n-1) x_i \text{ we have } \langle x_i, y \rangle > 0\}$. By previous argument, V contains a pair of antipodal points x & $-x$. So for at most $(2n-2) x_i$ we have $\langle x_i, x \rangle > 0$ or $\langle x_i, x \rangle < 0 \Leftrightarrow \langle x_i, x \rangle \neq 0$. $\Rightarrow$ for at least $(k+2) x_i$ we have $\langle x_i, x \rangle = 0$. But we can pick x_i in a GENERAL position s.t. this is impossible. $\times$.